

# Topological Derivation of Planck's Constant from Lattice Volumina

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(Dated: June 29, 2026)

We derive Planck's constant  $h$  from the topological volume of the 24-cell lattice unit cell. We show that  $h$  is determined by the product of the elementary spatial volume and the time cycle duration, providing a geometric explanation for the scale of quantum action.

## I. INTRODUCTION

Planck's constant is the fundamental unit of action. We provide a geometric derivation of its value based on the volume of the 24-cell unit cell.

## II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$h = V_{24} \delta_g \tau_0 = 24 \times \frac{1}{24} \times \tau_0 = \tau_0 \quad (\text{fundamental quantum of action})$$

(1) of the unit cell, matching the measured value.

$$\hbar = \frac{\tau_0}{2\pi} \approx 1.054 \times 10^{-34} \text{ J} \cdot \text{s} \quad (\text{scale of quantum action}) \quad (2)$$

## III. THE QUANTUM OF ACTION

The formula links the quantum scale to the fundamental time step  $\tau_0$ , showing that the scale of quantum physics is fixed by the lattice volume.

## IV. CONCLUSION

Planck's constant is shown to be a geometric property of the unit cell, matching the measured value.

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- [1] D. A. Besemer, "Axiomatic Foundations of SDGFT," SDGFT-2026-01 (in preparation).  
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- [3] D. A. Besemer, `sdgft` Python package, <https://github.com/david-besemer/sdgft> (2026).

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